

Technical Proposal: IR-Guided Docking Station

Sep 29, 2026

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1. Project overview

We will build a 3D-printed docking station that a mobile robot finds and connects to on its own, using infrared light to guide the last two metres.

Problem. Today the robot's batteries are taken out and charged by hand. The robot is only available if someone remembers to charge it, the battery handling takes time, and the robot has no fixed place to park.

Users. Robot operators are the main users: they should not have to think about charging. People who use the robot benefit because it is ready more often. Developers get a docking system they can reuse.

Expected value. The robot drives back to its station when needed, parks in the same place every time and makes a confirmed electrical connection. This removes routine battery handling and is the first step toward the robot charging itself overnight.

2. Functional requirements

The system must do the following:

1. The station sends out infrared signals that tell the robot whether it is to the left of the station, to the right, or straight in front.
2. The station sends a separate short-range signal that is only visible within about 30–40 cm, so the robot knows it is close.
3. The robot reads these signals with three infrared sensors on its front.
4. The robot steers itself onto the station's centre line and drives toward it.
5. When the robot is close, it lowers two metal contact rods, which are kept raised while it drives around.
6. A switch on the robot confirms that the rods are fully lowered.
7. The rods land on two spring-loaded metal rails on a floor pad in front of the station.
8. The robot detects when it has electrical contact with the station and stops.
9. The robot can back off the pad and raise its rods again.
10. Both the station and the robot log what they are doing over a serial connection, for testing.

3. Non-functional requirements

1. **Reliability.** The robot docks in at least 8 of 10 attempts.

2. **Starting area.** Docking works from up to 2 m in front of the station and up to 45° to either side.
3. **Accuracy.** Each contact rod must land on its rail. The rails allow about 2 cm of sideways error and 7–9 cm of overshoot.
4. **Lighting.** Docking works under normal indoor lighting.
5. **Safety.** The contact rods are raised whenever the robot drives, so they cannot catch on doorsteps, rugs or cables.
6. **Speed.** The robot moves slowly on the final approach so it can correct its path.
7. **Build.** The station and pad are 3D printed and use parts that are easy to buy.

4. MVP definition

The smallest working version is a robot that docks itself from the defined starting area and makes a confirmed electrical contact.

Included:

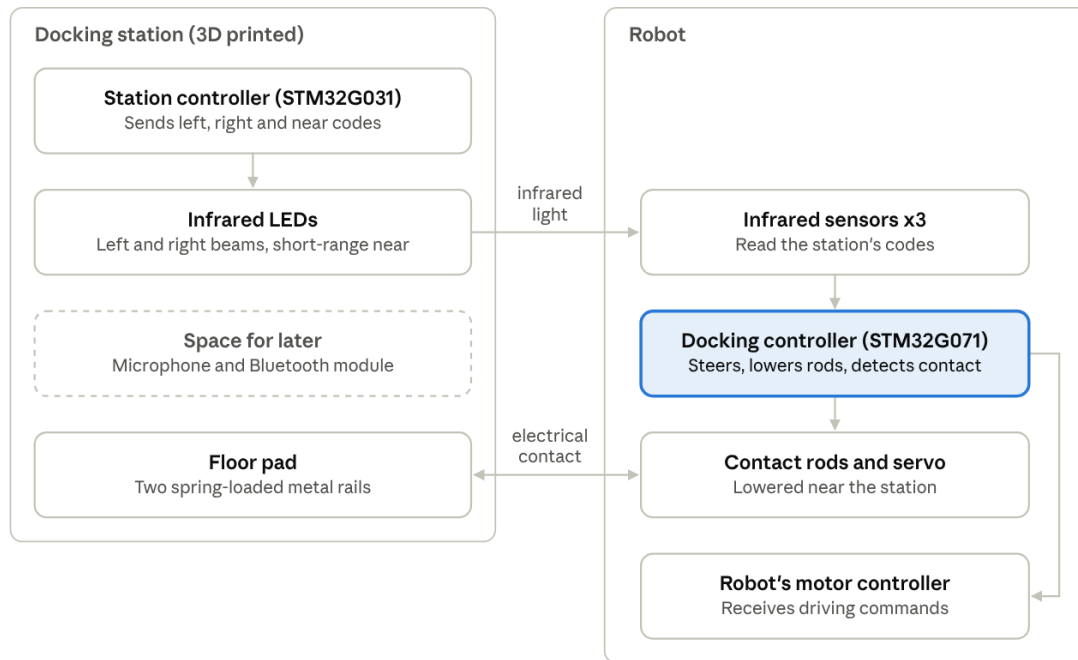
- The station with its infrared signals and the floor pad with rails.
- The robot's infrared sensors, lowering contact rods and the software that steers it in.
- Contact detection that tells the robot it has docked.

Success: 8 docks out of 10 attempts from different starting points within 2 m and 45° to either side, under normal indoor lighting.

5. System architecture

The system has two halves that work together without a data link: the station sends light, and the robot reads it and drives in.

The station light guides the robot in; rails and rods confirm the dock



Docking sequence:

1. The station's controller sends three coded infrared signals, one at a time: left, right and near. A thin wall between the left and right LEDs makes a sharp centre line.
2. The robot's three sensors pick up the codes. Seeing only left or only right tells the robot which way to turn; seeing both means it is on the centre line.
3. When the near signal appears, the robot slows down and the servo lowers the contact rods. A switch confirms they are down.
4. The robot creeps forward until the rods land on the rails. The docking controller senses the electrical contact and tells the motor controller to stop.
5. To leave, the robot backs off the pad and raises the rods, or the other way around.

The docking controller sends only simple driving commands to the robot's existing motor controller.

6. Technology selection

Each choice follows the same pattern: the options, what each costs us, what we chose and why. Our rule was the simplest proven solution that is accurate enough.

6.1 How the robot finds the station

Options: infrared light, a camera reading a printed marker, a laser distance sensor with a reflector, or radio positioning.

Trade-offs: Infrared is cheap and needs little computing power, but gives no distance. A camera gives full position but needs a computer and good lighting. A laser sensor is accurate but costs more. Radio positioning is only accurate to about 10 cm, which is too coarse.

Decision: infrared light.

Reason: it is accurate enough for the final approach, runs on a small microcontroller and fits the course scope.

6.2 How the infrared signal is sent and received

Options: a simple timer chip that blinks all LEDs the same way, or a microcontroller that sends a different code for each direction.

Trade-offs: A timer chip is simple, but every LED blinks identically, so the robot cannot tell left from right. A microcontroller can give each direction its own code.

Decision: the station's microcontroller sends a separate code for left, right and near, one at a time. The robot uses ready-made infrared receivers.

Reason: the receivers filter out lamps and daylight on their own and output clean digital codes, so no analog circuit tuning is needed.

6.3 Microcontrollers

Options: STM32 development boards, Arduino, or a timer chip only.

Trade-offs: Arduino is quick to start but has weaker timers. A timer chip cannot send different codes. STM32 takes a little more setup.

Decision: STM32 Nucleo boards: a small one (G031K8) for the station and a larger one (G071RB) for the robot.

Reason: both use the same tools, and their hardware timers generate the infrared signal without loading the processor. The station board also has spare pins for the future expansion, and share ecosystem with the existing robot hardware.

6.4 Contacts

Options: spring contacts on the front of the station that the robot pushes into, or contacts at floor level that the robot drives over.

Trade-offs: Front contacts need the robot to stop at an exact distance and push against springs. Floor-level contacts allow the robot to stop anywhere along the rail, but rods hanging under the robot could catch on things.

Decision: floor-level contacts. The station has two spring-loaded rails on a floor pad. The robot has two rods, about 10 cm apart, that a small servo motor lowers only near the station.

Reason: the robot does not need to hit an exact spot or push the station, and the rods stay out of the way while driving. A switch confirms the rods are down, and springs raise them if the servo loses power.

7. Project management plan

The work splits naturally into the station side and the robot side, with shared mechanical and integration work.

Santeri Karjalainen: robot-side software (reading the sensors, steering, lowering the rods, contact detection), connecting to the robot, and overall planning.

Zohaib Salman: station software (infrared signals) and wiring.

Shared: the contact rod mechanism, testing and documentation. The station and floor pad design and 3D printing.

How we work:

- We create a task list for the baseline structure of the project steps.
- Well structured GitHub repository to keep track of changes made.
- Discord for quick communication and sharing.
- Each part is tested on the bench before it is joined with the rest.
- We meet weekly to update the board and decide the next week's tasks.

8. GitHub repository

Repository URL: <https://github.com/skainen/docking-station>

One repository holds the station software, the robot software, the 3D models and the documents. The README describes the problem, how the system works, the main parts and the milestones.

9. Risk assessment

The biggest risks are stray infrared light and parts delays; both are dealt with in the first milestone.

Stray infrared light. Sunlight, lamps or reflections off walls could confuse the robot. *Impact:* high, since wrong signals mean failed docks. *Mitigation:* coded signals and receivers that filter out steady light, shades around the sensors, adjustable LED brightness, and early tests in the real room.

Parts delays. The university orders only from DigiKey's own stock, and some parts may be out of stock. *Impact:* medium, delays the first milestone. *Mitigation:* order in week one, pick parts DigiKey stocks itself.

Robot not available. *Impact:* high, blocks integration. *Mitigation:* develop the steering on a small test base first; the docking board talks to the robot through one simple connection.

Rods miss the rails. *Impact:* medium. *Mitigation:* wide rails, a sharp centre line from the station, slow final approach, and low guide rails for the wheels.

Contact mechanism wears or jams. *Impact:* medium, false or missed contact. *Mitigation:* a hard stop takes the load off the servo, a switch confirms the rods are down, springs raise them on power loss, and the software double-checks contact before stopping.

One team member unavailable. *Impact:* medium. *Mitigation:* shared documentation and reviewing each other's code keep both familiar with the whole system.

10. Development plan

Four milestones take the project from a working infrared signal on the bench to a full docking demonstration. Week numbers count from project start.

1. **M1 - Infrared signal (weeks 1-3).** Parts ordered in week one. The station sends its left, right and near codes, and the robot's sensors read them correctly on the bench. We test how room lighting affects the signal.
2. **M2 - Steering (weeks 3-5).** The docking software steers a test base onto the station's centre line from anywhere in the starting area.
3. **M3 - Contacts (weeks 5-7).** The station, floor pad and rod mechanism are printed and assembled. The rods lower on command, the switch confirms it, and contact is detected on the rails.
4. **M4 - Integration and demonstration (weeks 7-9).** Everything runs on the real robot. The final test is 10 docking attempts from different starting points, with results written up for the demonstration.